

Adobe Comments on the National Science Foundation's Request for Information on the Development of an Artificial Intelligence Action Plan

Introduction

Adobe welcomes the Administration's efforts to establish U.S. policy for sustaining and enhancing artificial intelligence ("AI") innovation in the U.S. in order to promote human flourishing, economic competitiveness, and national security, and we appreciate the opportunity to submit recommendations on the Administration's development of an AI Action Plan. Unlocking the full potential of AI will require a dynamic and flexible policy framework that enables the private sector to spur AI innovation in the U.S., while providing clear guidance to organizations on how they can leverage industry standards to inspire consumer confidence and accelerate adoption. By taking this approach, the U.S. can achieve the Administration's goal of continuing to lead the world on AI.

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Background

At Adobe, our mission is to change the world through personalized digital experiences. Since our founding in December 1982, we have pioneered transformative technologies that allow our customers—who range from emerging artists to global brands—to channel their imaginations, unleash their creativity, and power their businesses.

As technology evolves and becomes more advanced, we are committed to innovating responsibly. This means making sure our technologies and the processes we go through to develop them are accountable, responsible, and transparent so that we can rapidly innovate while assuring consumer confidence in our products. We believe that our approach to development strikes the right balance to promote human advancement and cement U.S. technological leadership and should serve as an industry standard for all technology companies.

Adobe's business is comprised of three cloud-based solutions: Creative Cloud, Document Cloud, and Experience Cloud. Across each cloud and in our products, Adobe is building on a decade-long legacy of AI innovation by leveraging the power of AI to deliver hundreds of intelligent capabilities. For example, in Creative Cloud, AI powers many creative functions including advanced image editing features in Photoshop, making it easier for everyone to tell their story with simpler and more intuitive tools. In Document Cloud, Adobe Acrobat's AI Assistant leverages AI technology to provide users with powerful document interaction capabilities such as the ability to generate summaries, answer questions, and provide insights across multiple document types. And as part of our Digital Experience offerings, Adobe's customers can use AI-driven features to deliver

relevant and meaningful insights and personalized digital experiences to the vast number of visitors on their websites.

In March 2023, Adobe launched [Firefly](#), our family of creative generative AI (“GenAI”) models that are both creator-focused and safe for commercial use. Starting with GenAI image generation, Firefly allows users to channel their creativity in ways they never imagined possible by simply typing in a prompt and generating images in seconds and is available in multiple Adobe applications like Photoshop, Illustrator, and Adobe Express. Adobe is also bringing Firefly GenAI capabilities to our audio and video tools and plans to continue to expand into other modalities such as 3D. To date, Firefly has been used to generate over 18 billion images across Adobe’s creative tools including Photoshop, Express, Illustrator and more.

We believe that AI done right will amplify human creativity and capabilities to new levels with deeper insights, accelerated task performance, and improved decision-making ability.

Executive Summary

Adobe supports the Administration’s goal of enhancing AI dominance and support for shared democratic values globally through private sector innovation to promote human flourishing, economic competitiveness, and national security. Our response and accompanying recommendations highlight the following themes:

- **Strengthen American Leadership in AI Technical Standards that Support Innovation and Protect Online Speech:** Adobe supports the advancement of U.S.-led open technical standards to enhance AI transparency, safeguard national security, and protect free speech by providing consumers with reliable tools to verify the provenance of digital content for themselves.
- **Promote a Risk-Based Approach to AI Development That Fosters Free Market AI Innovation:** Adobe supports a risk-based approach to AI development which allows innovation to rapidly accelerate by reducing unnecessary regulatory barriers and inspiring consumer confidence.
- **Unlock the United States’ AI Potential by Guaranteeing Long Term Access to High Quality AI Training Data:** We urge the Administration to take a pro-innovation approach to data access that removes regulatory uncertainty for AI developers and safeguards creators from commercial misuse.
- **Explore an AI Testing Pilot and Standardization Benchmarks:** We urge the Administration to promote AI innovation, while ensuring secure standards, by establishing an AI Testing Pilot and developing standardized test datasets, enabling companies to evaluate their models through a scalable, industry-driven framework that fosters public trust.
- **Unleash the Power of AI to Modernize the U.S. Federal Government:** Adobe is supportive of efforts to leverage AI to enhance efficiency, reduce costs, and improve public services across the Federal Government.

Strengthen American Leadership in AI Technical Standards that Support Innovation and

Protect Online Speech.

As referenced in recent [White House communications on enhancing America's AI leadership](#), the Administration has stated its intent to strengthen American leadership in AI technical standards. Adobe recommends that the U.S. move quickly and with intentionality to ensure continued global leadership in promoting and developing AI-related open technical standards to ensure that international regulatory regimes foster the creation of AI technologies, rather than strangle them. Understanding that U.S. leadership in global AI standards depends on the involvement of various stakeholders within this ecosystem, the Administration should look to further develop and promote private sector-led, U.S. AI standards globally. Our approach to accomplishing this goal aligns with the instructions provided in [Executive Order 14149, Restoring Freedom of Speech and Ending Federal Censorship](#), where the Administration can promote American solutions internationally that give consumers the tools they need to help make decisions about online content without interference or censorship from government or industry.

Methods to improve awareness and transparency regarding the origins of digital content is ripe for U.S.-led global standardization. This is particularly relevant in identifying whether content is AI-generated or AI-edited, along with more comprehensive information about the content's origins, history, and context. Thanks to advancements in AI, developing, editing, and distributing content has become more powerful and accessible for consumers. However, the same tools enhanced by the power of AI to make and share legitimate content can also be weaponized to undermine national security, and sow political and cultural discord. It is critical to protect the integrity of online speech by ensuring Americans have access to tools that verify the authenticity of digital content.

As the Administration formulates the AI Action Plan to enhance America's global AI dominance, we urge the government to leverage industry standards such as the consensus-based, interoperable standard pioneered by the [Coalition for Content Provenance and Authenticity \("C2PA"\)](#), which is now providing consumers with a simple, reliable method of determining the provenance and authenticity of the content they are consuming online. This technology, called [Content Credentials](#), provides a 'nutrition label' for content that can show information about images, videos, documents, audio files and more, such as how they were created, the creator's name, the date created, and edits that were made along the way.

With Content Credentials, creators can indicate not only whether GenAI was used, but how it was used. This level of transparency helps consumers to make more informed decisions about whether to trust the content they see online and helps ensure government and big tech never become the sole arbiters of truth. The Adobe-led [Content Authenticity Initiative \("CAI"\)](#) is a global, cross industry coalition, supported by over 4,000 members, working to advance free, open-source implementations of Content Credentials' provenance technology. At Adobe, we have promoted this level of transparency by integrating Content Credentials into popular products such as Photoshop, Lightroom and Firefly.

By driving adoption of industry standards like the C2PA and promoting technologies like Content Credentials, the U.S. can set a global benchmark for AI transparency and help keep the government and big tech out of the censorship business. To achieve these goals, we recommend the following actions:

- **Set the Global Standard for Content Provenance with the C2PA.** We recommend that the Administration leverage its convening authority and work through international bodies, to which the U.S. is a party, to advance the development and adoption of existing open technical standards for digital content provenance (i.e. C2PA). The C2PA is currently working with the International Organization for Standardization's ("ISO") TC 171/SC 2 technical committee, whose charter includes content authenticity, to establish Content Credentials as an open standard from a recognized, international standards developing organization.
- **Adopt Content Credentials to Foster Trust in Government Content.** To strengthen public confidence in the integrity of the U.S. Government's digital content, we urge the White House to issue guidance directing government agencies to implement the C2PA standard and Content Credentials for all government digital content, such as images, videos, audio, and document content they post on official government websites and communications. The broad adoption of the C2PA standard would enable the government to share trusted resources and information with the American public and help prevent consumers from falling victim to harmful deepfakes or synthetic content spread by bad actors. We recommend the White House stand up a federal interagency task force to promote the inspection of the C2PA standard and global adoption of Content Credentials by the U.S. government and its partners.
- **Protect the Authenticity of Online Speech and Protect Consumers from Censorship by Ensuring Online Platforms Maintain and Display Content Credentials.** For Content Credentials to be maximally effective, they must reach consumers in all the places where they consume information, such as major internet platforms. Currently, Content Credentials are often stripped away on internet platforms, depriving consumers of critical information and context about the content they are consuming. We urge the Administration to require online platforms to maintain and display any Content Credentials present in digital content on their systems. If a piece of content has Content Credentials attached, those Content Credentials should not be stripped away. With access to the important context that Content Credentials provide, we can empower consumers with a free-market solution to make their own decisions about what to trust online, rather than being subject to the determinations of the platforms where content is hosted.
- **Encourage Smartphones to Implement Content Provenance.** Provenance must be integrated at every stage of the content lifecycle, from point of capture to editing to publication. This also includes the devices most commonly used today for capturing digital content, such as smartphones. We have already seen significant progress around implementing the C2PA open technical standard for Content Credentials into cameras.

Now it's time for smartphone manufacturers to follow suit. In order for this solution to be maximally effective we need to ensure this technology is available wherever consumers capture content, which of course includes smartphones. We urge the Administration to encourage smartphone manufacturers to implement provenance capabilities, so that users have the option to attach content provenance metadata to all newly created and captured content.

Promote a Risk-Based Approach to AI Development That Fosters Free Market AI Innovation.

As the Administration considers its approach to reducing regulatory barriers to promote AI innovation as set forth in [Executive Order 14179](#), Adobe encourages consideration of the adoption of a risk-based approach to evaluating AI that enables companies to rapidly innovate while ensuring consumer confidence in AI technologies. Adobe envisions a world where the most promising AI is developed, with widespread adoption, making us all more prosperous and more productive.

Countries around the world are looking to regulate AI. We have seen varying approaches to AI regulation such as analyzing AI model size, scrutinizing potential catastrophic safety implications, and implementing onerous, and trade secret-compromising, training data transparency requirements. None of these approaches achieve the goal of spurring innovation and driving widespread adoption of AI by consumers. To balance these objectives and ensure global leadership in AI, it is critical that the U.S. government promote policies that enable companies to go to market faster while making sure that consumers feel comfortable and confident in using AI. These policies, done right, will avoid an overly precautionary regulatory regime, while ensuring that all Americans benefit from AI technology. This is an opportunity for the Trump Administration to establish a framework for AI governance that can and should be adopted globally.

Adobe believes that a risk-based approach achieves these goals, and we urge the U.S. government to consider Adobe's approach to AI development as a framework that should be encouraged globally.

Beginning in 2019, Adobe proactively and voluntarily developed a layered, multi-disciplinary process for responsible innovation to enable the company to rapidly develop AI while ensuring the confidence of our users to take advantage of this new technology. This development process included establishing an AI governance program with a comprehensive review process that includes an impact assessment, ensuring models are free from ideological bias. As such, we have suggested the following key elements that we believe should be incorporated into the Administration's AI Action Plan to promote a risk-based approach to AI development:

- **Promote a Risk-based Approach to Foster Innovation.** Establishing a risk-based approach to AI governance that focuses on only the most high-risk use cases will foster innovation with the added benefit of requiring organizations to conduct a more rigorous and thorough review of high-risk AI systems. At Adobe, we have a multipart review process

with an integrated AI impact assessment that is designed to assess the impact of an AI product or feature, including mitigation of any ideological bias. If an initial assessment shows the AI product or feature is low-risk and the feature meets our internal standards, the feature is approved and goes to market. For example, an AI feature that recommends a set of fonts based on a document template would move through a faster review process. AI features that have a higher potential impact, like the possibility of displaying an ideological bias, go through a more rigorous testing process, including a review by the AI Review Board. A risk-based approach like this will be critical moving forward as AI continues to advance and the United States works to promote innovation, while also ensuring that consumers have confidence in the AI technology they utilize.

- **Encourage Companies to Solicit Ongoing Feedback.** Community feedback can help mitigate potential biased, inaccurate and harmful outputs of AI systems and equip consumers to trust the AI that they are using. At Adobe, when we launch new generative AI features, we generally do so first in a beta format to seek input from our employees, customers, and broader creative community. In addition, we design our generative AI features (like those powered by Adobe Firefly) with a built-in feedback mechanism so users can easily report if a feature produces a result they perceive as biased or inaccurate. This feedback data allows our product and engineering teams to help identify harmful outcomes and address them appropriately. In our view, this constant feedback loop with our user community is the best way to help ensure our tools minimize bias and produce accurate results. A feedback mechanism creates transparency for our customers and eliminates the need for government to intervene and require onerous reporting requirements that risk compromising trade secrets.
- **Leverage Public-Private Partnerships.** Over the past several years, AI development has occurred at a rapid pace across various sectors of industry. Many companies, including Adobe, have garnered valuable experience and knowledge in navigating the rapid growth of this cutting-edge technology and mitigating against potential harms. We urge the Administration to leverage industry learnings and industry collaboration as they continue to identify ways to accelerate AI innovation in the U.S.

Unlock the United States' AI Potential by Guaranteeing Long Term Access to High Quality AI Training Data.

As stated in [Executive Order 14179, Removing Barriers to American Leadership in Artificial Intelligence](#), it is the policy of the United States to sustain and enhance America's global AI dominance in order to promote human flourishing, economic competitiveness, and national security. With the right Government policies, we can solidify our position as the global leader in AI and secure a brighter future for all Americans. We believe providing AI systems developed by Americans will need decades of high-quality data to meet these objectives. We would like to offer a few policy recommendations that will help the United States lead the world in developing AI systems based on the availability of high-quality training data.

The Administration has a world-leading opportunity to establish a data access framework that fosters AI innovation, while incentivizing creativity for decades to come. Other nations are actively considering opt-out standards for AI training data, and the U.S. has a chance to set a global precedent. By implementing a long-term approach, the U.S. can ensure that companies of all sizes—including startups—have access to the data needed to build competitive AI models, while also giving those who create training data some fundamental safeguards from their work being commercially misused.

In order to accomplish these goals, the U.S. should consider the following policy positions:

- **Be Declarative on Copyright Fair Use of Training Data.** In the absence of clear regulatory guidance on whether the use of data to train AI is considered fair use under copyright law, we encourage the Administration to take a bold, pro-innovation stance and be declarative on the issue, giving American AI developers of all sizes a clear framework and certainty in the marketplace as AI systems are developed.
- **Facilitate an Opt-Out Standard.** While it is essential that government establish a framework that enables broad access to data, it is also important to recognize that there is no clear, globally accepted, standard for creatives to opt-out of having their works used to train AI models. Therefore, the Administration should, in parallel, advance an opt-out standard that enables creators to indicate this preference for specific pieces of content. As explained in greater detail below, the global open standard around [Content Credentials](#) can be leveraged to meet this need; creators can use Content Credentials to attach a “do not train” tag to an individual piece of content that remains associated with the work wherever it goes. Therefore, Content Credentials can provide a tamper-evident way to help ensure that an individual’s AI training preferences are respected, and their content is treated the same across all sites where they may choose to publish it.
- **Establish Anti-Impersonation Rights for Creators.** As the Administration considers some of the important questions raised by GenAI, we believe it should consider supporting a new specific protection for creators that copyright law currently doesn’t cover. One of the core questions raised by Adobe’s customers is, what happens when someone uses an AI model to replicate their style, in direct economic competition with their original work? Adobe believes the intentional misuse of AI tools in this way could pose a legitimate issue for the creative community. In response, Adobe has proposed establishing a new anti-impersonation right to address this type of economic harm in the U.S. Such a law would provide a narrow ability for a person to protect their most valuable assets from being intentionally and commercially used to impersonate their work and style through AI tools. This right would provide a new mechanism for artists to protect their livelihood from people misusing this new technology without having to rely solely on mismatched existing laws around copyright. Paired with this protection, the U.S. can implement a framework that broadly gives more access to data while minimizing the risk of AI harms, therefore unlocking an abundant supply of AI training data available to AI developers for decades to come.

The combination of the recommendations above will ensure the U.S. solidifies its position as the global leader in AI through the abundant supply of AI training data, securing a brighter future for all Americans.

Explore an AI Testing Pilot and Standardization Benchmarks.

Adobe recognizes the importance of the Administration's goal to develop AI systems that are free from ideological bias and engineered social agendas as stated in [Executive Order 14179](#). That is why we have established a comprehensive AI Governance Program that includes rigorous testing, training, and oversight by our AI Review Board. Rather than imposing burdensome government requirements that could restrict private sector AI development and deployment, the Administration has an opportunity to take an alternative regulatory approach, unique to the U.S., by establishing robustness, resilience, and safety standards and an AI Testing Pilot to ensure that users, including younger users, can interact safely and appropriately with AI systems. The AI Testing Pilot would include developing robust benchmark test datasets that companies can use to evaluate their AI models.

Standards provide consistent guidelines to ensure quality, safety, and interoperability for AI systems, ensuring consistency across different industries. At the same time, standards encourage responsible development, that allows companies to build trust with their customers, by giving companies clear guidelines without imposing unnecessary restrictions.

The Federal Government is best situated to convene the multitude of stakeholders and perspectives needed to develop these standards and create benchmarking, testing systems that companies can test their AI models against to ensure they meet these standards. The benchmarks would be focused on allowing companies to push the boundaries of AI technology without unnecessary restrictions, while ensuring that users—especially parents—have the right tools and information to make informed decisions about which AI systems to interact with or let their children interact with. The benchmarks could address categories such as violence, sexual content, and self-harm.

In addition, the Federal Government could establish a classification system to communicate the level of adherence to these standards and benchmarks. This will help users and stakeholders make informed decisions about content and technology and whether it is suitable for specific applications. By highlighting potential concerns, such as explicit content or AI bias, classification systems help protect vulnerable audiences, including children. This approach preserves innovation and freedom, allowing new ideas to flourish while maintaining public trust.

To ensure consistency and maximum benefit, we propose:

- **Partner with the Private Sector to Develop Benchmark Test Datasets.** Encouraging collaboration between the private sector and federal agencies in building standards, classification systems, and benchmark test datasets would create a regulatory

framework that prioritizes both innovation and responsibility. This would help create an environment for a scalable and industry-driven approach that fosters consumer trust without hampering technological progress. Rather than burdening AI companies with restrictive compliance measures, this approach would incentivize them to refine their models through transparent, standards-driven evaluation, ensuring AI development remains innovative, effective, and globally competitive. Benchmark test datasets would need to be established across various verticals, industries, markets, and use cases. There is an opportunity for the private sector to partner with agencies to establish appropriate datasets in each case. Benchmark test datasets may also open the door to a self-certification regime of compliance, which would, in turn, build greater consumer trust.

- **Establish a Self-Certification Process for Companies.** By establishing a self-certification process for AI systems, the Government can set the norms for the robustness, resilience, and security standards of AI models. Companies are accountable for their development of AI and have an incentive to participate, as it creates a safe harbor against future litigation. Companies can use third-party auditors to verify that they meet benchmarks, creating legal certainty for companies and assuring governments and consumers that AI was soundly developed. For example, together with industry, government could establish certain acceptable ranges of outcomes for sexual content for different use cases of AI models. These outcomes could be established as standards that companies strive to meet depending on the use case of their specific AI model. Then, government and industry work together to create testing and evaluation datasets that companies can run their AI models against to determine whether they fall into the standard range and do further testing and refining if needed. Then, a company could self-certify that they have met the accepted standard. This will not only allow companies to innovate and test in a more efficient and secure way, but will also create much needed consistency, assurance and clear consumer understanding of the AI they are interacting with.

Unleash the Power of AI to Modernize the U.S. Federal Government.

[Executive Order 14158](#) establishes the President's Department of Government Efficiency (DOGE) and instructs the DOGE Service Administrator to commence a Software Modernization Initiative to improve the quality and efficiency of government-wide software, network infrastructure, and information technology (IT) systems. As such, Adobe encourages the Administration to utilize AI tools to assist with the required modernization of Federal technology and software to maximize government efficiency and productivity. AI can be adopted and leveraged to meet the Executive Order's requirements by enhancing efficiency, reducing costs, and improving public services across the Federal Government. The Administration also has the opportunity to expedite the acquisition of AI solutions and unleash untapped AI capabilities within existing software, already utilized by agencies, that meet [Executive Order 14179's](#) standard of being free of ideological bias or engineered social agendas. AI systems produced by American companies, especially those that are commercially safe or trained only on licensed content, or content where permission is not required by law, are prime for this express

acquisition process.

As a result of rapid advancements in AI, the U.S. Government has a tremendous opportunity to implement AI-driven solutions in areas such as digital service delivery and content creation, streamlining operations and providing more accurate and timely responses to public needs. By adopting AI technologies, the government can better manage resources, predict and respond to challenges, and deliver higher quality services to its citizens. GenAI will enable federal employees to be more productive and increase public access to federal services.

Below are several recommendations where the federal government could take advantage of GenAI and machine learning capabilities to improve agency missions and business operations, while simultaneously meeting the Administration's goal of modernizing legacy software and increasing efficiency:

- **Unlock Existing AI Tools Within Government Software.** Many secure, U.S.-Sourced AI tools are already built into software utilized by Federal employees but are inaccessible to them due to overly complex and often distorted usage guidelines. These tools, if unlocked, will empower federal workers to automate routine tasks, rapidly analyze vast datasets, and make data-driven decisions, significantly boosting productivity and enabling them to focus on high-value initiatives.
- **Streamline Procurement of Commercially Safe AI Systems.** The Administration released [Executive Order 13859](#), which encourages federal agencies to reduce barriers to government acquisition and use of AI. Prioritizing procurement of AI tools can serve to increase federal worker's productivity and maximize government efficiency. Commercially safe AI tools, those that train only on licensed content, or content where permission is not required by law, are ripe for expedited procurement, as they mitigate some concerns posed by other AI systems.
- **Use AI to Optimize Information Processing and Data Analysis.** Government employees can leverage GenAI systems to revolutionize information management and data analysis. AI-powered tools can rapidly search and simplify information from complex documents, extract insights from multiple disjointed internal sources, and automate the summarization and reporting of external requests from citizens. By implementing industry-leading AI tools, government agencies can significantly enhance their efficiency in handling the vast amounts of information they receive, improve decision-making processes, and ultimately deliver more responsive and informed services to citizens, all with less burden placed on Federal workers.
- **Create and Deliver Content More Efficiently Using AI.** By leveraging AI tools, the Federal Government will be able to train non-technical communicators to create images or videos that leverage pre-approved templates, logos, or style guides, to quickly generate informational content to be distributed within minutes across digital touch points. This reduces dependencies on IT staff to hard code content while enabling communications

and marketing teams to respond in real time. Government can also pair generated content with a content management system, allowing organizations to generate and manage AI content from creation to publication to deletion.

As AI development continues to accelerate, the U.S. Government should work to implement AI-driven solutions in digital service delivery to optimize their operations and better address federal worker and public needs. Initiatives to do so, such as unlocking existing AI systems, are key to not only meet the requirements laid out in [Executive Order 14158](#), but also to enhance government efficiency, reduce agency costs, and improve public services across the Federal Government.

Conclusion

Adobe appreciates the opportunity to provide feedback on the Administration's development of the AI Action Plan. We appreciate that the National Science Foundation has taken important steps to consult with interested stakeholders as the Administration formulates its AI policy priorities.

We look forward to continuing the discussion with the Office of Science and Technology Policy and the National Science Foundation and are eager to engage collaboratively on these important issues.